PROBLEM STATEMENT :

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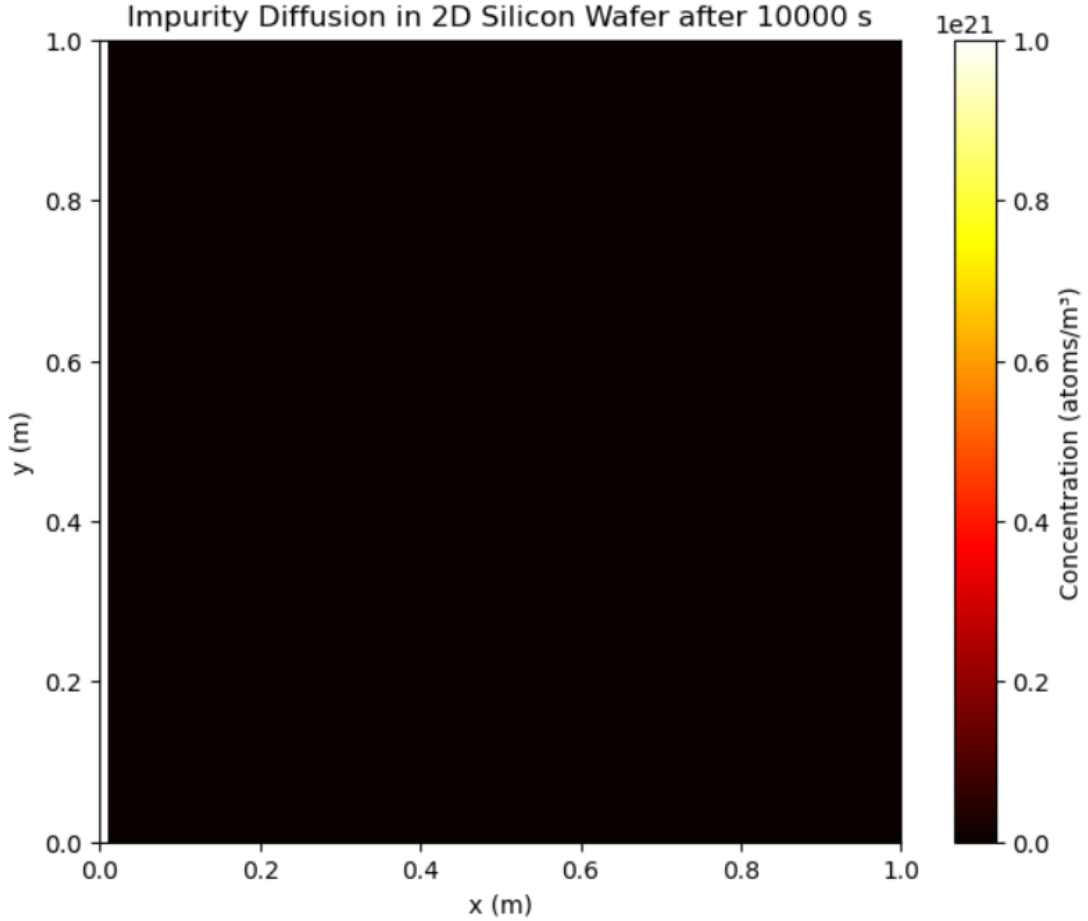
```python
import numpy as np
import matplotlib.pyplot as plt

# Constants
L = 1.0                  # length of the wafer in meters
N = 100                  # number of grid points in each direction
dx = L / (N - 1)         # grid spacing
D = 1e-12                # diffusion coefficient in m²/s
C0 = 1e21                # initial concentration at left edge
t_total = 10000          # total simulation time in seconds
dt = 0.25 * dx**2 / D    # time step (stability condition for explicit method)
steps = int(t_total / dt)

# Initialize concentration grid
C = np.zeros((N, N))
C_new = np.zeros_like(C)

# Apply initial condition: left edge = C0
C[:, 0] = C0

# Time evolution
for _ in range(steps):
    # Update using 2D explicit finite difference
    for i in range(1, N-1):
        for j in range(1, N-1):
            C_new[i, j] = C[i, j] + D * dt / dx**2 * (
                C[i+1, j] + C[i-1, j] + C[i, j+1] + C[i, j-1] - 4 * C[i, j]
            )
```

```python
    # Reapply boundary condition (left edge constant)
    C_new[:, 0] = C0
    C_new[:, -1] = C[:, -1]   # Right edge: Neumann (no flux)
    C_new[0, :] = C[0, :]     # Top edge: Neumann
    C_new[-1, :] = C[-1, :]   # Bottom edge: Neumann

    # Update concentration
    C, C_new = C_new, C

# Plotting the result
plt.figure(figsize=(8, 6))
plt.imshow(C, extent=[0, L, 0, L], origin='lower', cmap='hot')
plt.colorbar(label='Concentration (atoms/m³)')
plt.title(f'Impurity Diffusion in 2D Silicon Wafer after {t_total} s')
plt.xlabel('x (Time)')
plt.ylabel('y (Concentration)')
plt.show()
```


Impurity Diffusion in 2D Silicon Wafer after 10000 s
1e21
1.0
0.8
0.6
0.4
0.2
0.0
y (m)
0.0
0.2
0.4
0.6
0.8
1.0
x (m)
Concentration (atoms/m³)